

Attention-Based Generalization for Diabetic Retinopathy: A Comparative Study of Hybrid and Standard Deep Learning Architectures

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Abstract—Diabetic Retinopathy (DR) is the primary cause of vision impairment and blindness among diabetic patients. Making an early and accurate detection is essential for effective treatment. In recent years, deep learning based approaches are having a great potential in automated DR classification. Despite significant advancements in the detection of DR, few challenges like class imbalance, overfitting and limited generalization remain as key concerns. This study presents a comprehensive approach for multi-class DR classification using deep convolution neural networks. A comparative study and analysis of EfficientNetB0, ResNet50, MobileNetV2 is conducted under consistent experimental conditions to understand and evaluate their performances. In addition, a hybrid architecture combining EfficientNet and ResNet is implemented, incorporating the attention mechanism to maximize the feature extraction and enhance feature representation which improves classification performance. Experimental results demonstrate that the hybrid model achieves higher validation accuracy because of enhanced feature learning. However, EfficientNetB0 exhibits superior generalization which reflected in the values of evaluation metrics, i.e., F1-score, recall, precision in domination compared to other implemented models. The findings highlight the importance of evaluating the models beyond accuracy alone and emphasize the behavior between the model complexity and generalization capability in medical image classification tasks. Overall, this work provides detailed comparison and valuable insights into the design of robust deep learning models of DR detection and contributes towards the development of reliable automated screening systems for real-world applications.

Keywords—Diabetic Retinopathy, Retinal Fundus Imaging, Transfer Learning, Hybrid Neural Network, Feature Fusion, Attention Mechanism

I. INTRODUCTION

Diabetic Retinopathy (DR) is an eye disease caused by the complication of diabetes [1]. DR is the most common cause of blindness among diabetic patients [2]. Diabetic retinopathy often does not have any symptoms in the early stages, it will start showing symptoms like blurry vision, colors appearing faded or washed out, more floaters when the disease has gotten worse [1]. So it is important to detect DR at the early stage for better, effective recovery and preventing irreversible vision loss. According to the World Health Organization (WHO), approximately 93 to 103 million people worldwide are living

with some form of DR and the number of people with DR is expected to rise from 126.6 million in 2010 to 191 million by 2030 which is a 150% increase in the number of individuals affected with DR [3].

The traditional and manual screening of DR consists of retina imaging, fundus photography, and optical coherence tomography but it has some limitations, like high cost for screening of each patient, highly time consuming and technical complexity [4]. The traditional method of analysis requires an expert ophthalmologist which is not scalable and not easily accessible all the time. So manual methods of screening is insufficient for large scale screening.

Artificial intelligence is transforming the detection of DR, moving away from traditional machine learning models that depends on manually created features to deep learning methods like Deep Neural Networks (DNNs) and Convolution Neural Networks (CNNs) that allow for automatic feature extraction from the uploaded retinal images [4]. Despite using the neural networks in the medical field of diabetic retinopathy detection there are still few major challenges to be addressed in the process of developing a deep learning model to detect the DR, which involves — multiclass classification, dataset imbalance, generalization issues and overfitting.

This study proposes a comparative analysis of multiple deep learning architectures and neural networks to address the challenges of existing approaches. Transfer learning based models like EfficientNet, ResNet, MobileNet are analyzed and compared due to their effectiveness in image classification and object recognition. Also a hybrid architecture combining the EfficientNet and ResNet is also proposed to understand the performance of the hybrid architectures in detection of the diabetic retinopathy.

The main contributions of this work are summarized as follows: (i) a detailed comparative analysis of various deep learning architectures — EfficientNet, ResNet, MobileNet for multi-class diabetic retinopathy classification; (ii) the integration of attention mechanism to improve and enhance feature selection and improve classification accuracy; (iii) the application of class weighting techniques to address class

imbalance issue in the dataset; (iv) the implementation of a hybrid EfficientNet – ResNet model that combines feature representations through feature level fusion; (v) an in depth evaluation of model generalization through the analysis of validation and evaluation metrics across different architectures.

II. RELATED WORK

In recent times DR scanning has completely shifted from manual scanning to automated approach using DL methodologies. This shift of diagnosis is mainly due to the huge population and many new patients affecting every year. Because of this the manual scanning can't support the needs of all the patients and hence automated based DL aided scanning for DR has swiftly taken over the medical industry. Most of the recent studies on diabetic retinopathy using DL have the following common points: (i) Even in 2023–2025 most of the researches in this has been still continuing with the help of CNNs like ResNet/DenseNet etc; (ii) From 2024 onward, decent number of papers has explored Vision Transformers (ViTs) for DR detection and grading. Several studies report competitive results compared with conventional CNNs.

A. CNN Based DR Detection

One of the key strengths of CNN based approaches is their ability to perform automated feature extraction unlike traditional methods that rely on manually designed features, which is important to their success in medical image analysis. CNNs learn features directly from data through a hierarchy of layers. In early layers, the network captures low-level features such as edges, color gradients, and simple textures, intermediate layers learn more complex structures such as blood vessel patterns and local texture variations, deeper layers identify high-level, clinically relevant features such as microaneurysms, hemorrhages which are key indicators of diabetic retinopathy. This hierarchical learning process is carried out using convolution operations, where filters are improved during the training to detect any meaningful patterns [5].

Various studies have achieved promising results and accuracy varying around 85–95% [5], [6]. The major limitations specified in the studies using this approach are: (i) the generalizability of the data is very limited and the results are significantly lower when compared to the balanced datasets; (ii) the CNN based models require a large and diverse dataset for achieving good results [5].

B. Transfer Learning Based DR Detection

Transfer learning in DR detection involves taking a CNN model such as EfficientNet, ResNet that has already been trained on a large dataset like ImageNet and reusing its learned weights for retinal image analysis. Instead of training a deep network from scratch, the pretrained EfficientNet model is adapted to the DR dataset and the EfficientNet achieves strong performance in DR classification with good generalizability across severity levels [6]. Transfer learning reduces training time and resource consumption, and overall

cost making it suitable for real-world and resource constrained environments [7]. Recent studies have found that, EfficientNet based transfer learning used as a first-stage feature extractor, followed by classification or ensemble learning for final prediction enhances feature selection and classification performance [8].

C. Hybrid Models

Hybrid models integrate multiple deep learning models into a unified system to improve both feature representation and classification performance. Feature extraction in hybrid DR models is multi-level and complementary, which is the key reason for their superior performance.

In EfficientNet – ResNet hybrid systems, retinal fundus images are first passed through multiple backend networks, where each network extracts different types of features. EfficientNet captures fine-grained, high-resolution features due to its compound scaling, while ResNet captures deep semantic features through residual connections that allow very deep architectures to train effectively. The extracted feature maps are then either concatenated or fused, sometimes enhanced using attention mechanisms like coordinate attention [9]. Combining multiple CNNs enhances feature learning and achieves high classification accuracy and very good evaluation parametric values [10]. Hybrid models integrating CNN feature extraction with ensemble classifiers significantly improve performance and generalizability [11].

D. Limitations of Existing Work

Despite having remarkable improvements in the DR detection through CNN based, transfer learning and hybrid approaches, there are still some limitations that exists in the field of research on retinopathy detection. One of the primary limitations is the lack of comprehensive comparative analysis across multiple deep learning architectures. Many existing studies focus on the performance of a single model, making it difficult to evaluate their relative effectiveness under consistent experimental conditions.

Another critical limitation is the lack of importance on model generalization. A considerable number of studies report the performance metrics solely based on the validation accuracy without evaluating the models on the independent test datasets.

Class imbalance is also another important limitation that is inefficiently addressed. Diabetic retinopathy datasets typically contain a disproportionately higher number of normal or mild cases compared to severe stages. Many existing studies have failed to address this issue and incorporate effective strategies to handle this imbalance, leading to biased models.

III. PROPOSED METHODOLOGY

A. Dataset Description

The dataset used in this study is the APTOS 2019 Blindness Detection dataset [12] sourced from kaggle. This dataset mainly consists of retinal fundal images that are categorized into groups based on the severity of the DR. The classification

is done into 5 categories named as — No DR, Mild, Moderate, Severe, and Proliferative DR. A key challenge associated with the dataset is class imbalance, where a large number of images belong to the normal or mild categories, while fewer samples represent severe stages of the disease. This imbalance poses difficulties in training robust models, as it can lead to biased predictions favoring majority classes. Additionally challenges like variations in image quality, retinal features consists in the dataset which further increases the complexity of classification task.

B. Data Preprocessing

In advance to model training, all retinal fundus images is subjected to a series of preprocessing steps to ensure consistency with deep learning architectures. The input images are resized to a fixed resolution of 224×224 pixels to match the input requirements of the pretrained convolutional neural networks. This standardization enables efficient batch processing and uniform feature extraction across all models.

In addition, specific normalization techniques were applied for corresponding models to align with the preprocessing requirements of different architectures. EfficientNet, MobileNet and ResNet models require different input scaling methods due to differences in their pretrained configurations. To address this, appropriate preprocessing functions are applied for each model individually. For the proposed hybrid architecture, a combined preprocessing strategy was implemented to balance the input representations for both EfficientNet and ResNet branches.

C. Model Architecture

1) *EfficientNetB0*: EfficientNetB0 is a convolution neural network architecture that utilizes a compound scaling method to uniformly expand the network depth, width, and resolution. This balanced approach enables the model to achieve high accuracy with relatively fewer parameters compared to traditional CNN architectures. In our study, EfficientNetB0 is employed as a baseline model due to its strong capability in capturing fine-grained features from retinal images, such as microaneurysms and subtle texture variations. The model is initialized with pretrained weights from the ImageNet dataset, allowing it to leverage previously learned visual representations and adapt them to the diabetic retinopathy classification task through transfer learning. To optimize performance, finetuning is applied by unfreezing the deeper layers of the network, enabling the model to learn domain-specific features while retaining general visual knowledge from pretraining.

2) *ResNet50*: ResNet50 is a deep convolutional neural network, which helps mitigate the vanishing gradient problem in very deep networks. In this work, ResNet50 is utilized as a comparative model due to its ability to capture high-level semantic features from retinal images. Similar to EfficientNetB0, the model is initialized with pretrained weights and fine-tuned on the target dataset. Its deep architecture enables effective learning of complex retinal structures and disease patterns associated with different stages of diabetic retinopathy.

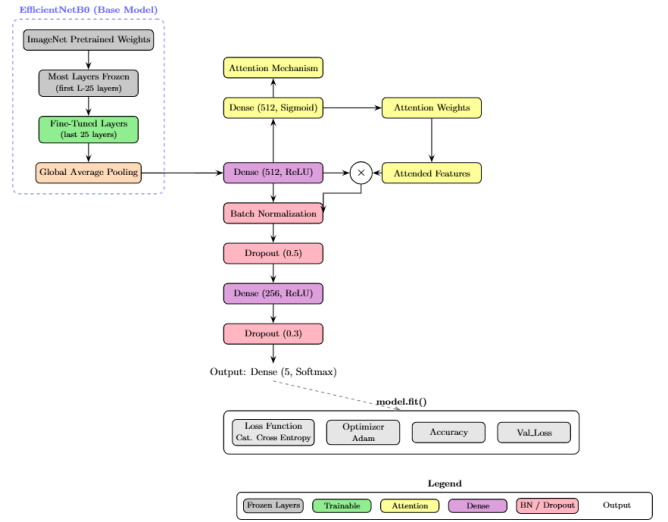


Fig. 1. EfficientNetB0 pipeline.

3) *MobileNetV2*: MobileNetV2 is a lightweight convolutional neural network architecture designed for efficiency in resource-constrained environments. It employs inverted residuals and linear bottleneck layers to reduce computational cost while maintaining competitive performance. In this study, MobileNetV2 is included as a comparative baseline to evaluate whether lightweight architectures can generalize effectively for multi-class DR classification. Like the other models, it is initialized with pretrained ImageNet weights and fine-tuned on the APTOS 2019 dataset.

4) *Proposed Hybrid Model*: Two variants of the hybrid model are implemented and evaluated in this study. Hybrid1 represents the initial hybrid configuration, where the last 40 layers of both EfficientNetB0 and ResNet50 are fine-tuned. It uses a larger classifier consisting of Dense(512), Attention(512), and Dense(256) layers, with a dropout of 0.5 and a learning rate of 1×10^{-4} . This configuration is focused mainly on maximizing feature learning capacity.

Hybrid2 is an optimized version designed specifically to reduce overfitting and improve generalization. The fine-tuning depth is reduced to the last 30 layers to prevent over-adaptation to the training data. The classifier complexity is also reduced to Dense(256), Attention(256), and Dense(128) layers. Regularization is strengthened by increasing dropout from 0.5 to 0.6, and the learning rate is reduced from 1×10^{-4} to 5×10^{-5} . Additionally, a fixed random seed is applied to ensure reproducibility across experiments. To leverage the complementary strengths of EfficientNetB0 and ResNet50, a hybrid architecture is proposed that integrates both models through feature-level fusion. The input retinal images are simultaneously processed by both networks, where EfficientNetB0 captures fine-grained local features, while ResNet50 extracts high-level semantic representations [12]. The feature maps obtained from both networks are transformed into feature vectors using global average pooling and then combining using

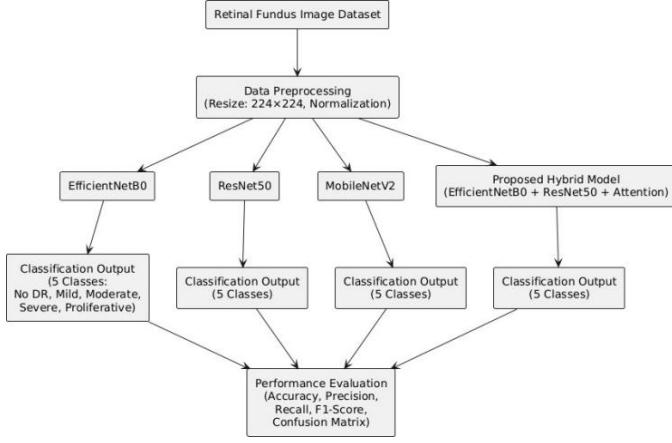


Fig. 2. Overview of System Architecture.

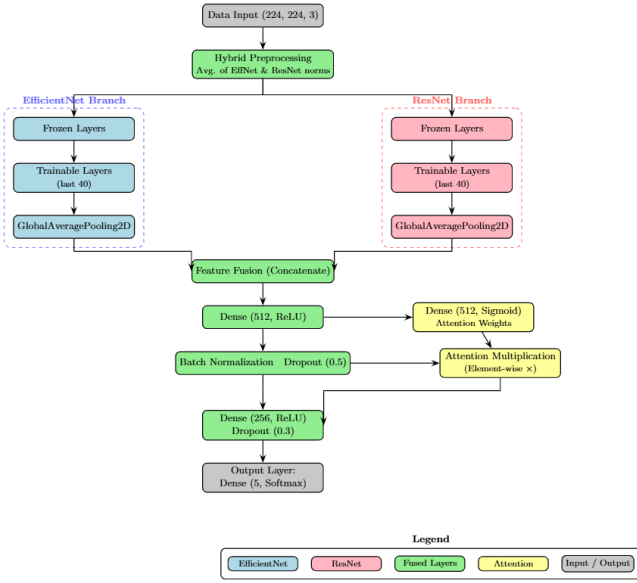


Fig. 3. Hybrid model Pipeline.

concatenation. This fusion strategy preserves information from both branches, enabling the model to learn a richer and more discriminative representation of retinal features. To further enhance feature selection, an attention mechanism is integrated after the fusion layer. The attention layer assigns adaptive weights to the fused features, allowing the model to focus on the most relevant patterns while suppressing less informative features. This improves the model's ability to distinguish between different stages of diabetic retinopathy. Additionally, fully connected layers with dropout regularization are employed to reduce overfitting and improve generalization [13]. The final classification is performed using a softmax layer that outputs probabilities across five DR severity classes.

D. Class Imbalance Handling

Class imbalance is a big problem in diabetic retinopathy datasets. This means that the samples are not evenly spread

out across different severity levels. Most of the images are in the normal or mild categories, and only a few show severe stages of the disease. This imbalance can cause biased model training, which means that the model tends to favor the majority classes and does poorly on the minority classes that are important for clinical use. To fix this problem, a strategy for weighting classes is used during model training [14]. In this method, classes that are not well represented are given more weight, which makes them more important to the loss function. This makes sure that the model learns features related to all severity levels better by punishing it more for misclassifying minority classes [14]. The use of class weights improves the model's ability to detect less frequent but critical stages of diabetic retinopathy, thereby enhancing overall classification performance and making the model more suitable for real-world medical applications.

E. Training Strategy

The models learn by using a transfer learning method, which means that pretrained weights from large image datasets are adjusted to work better with the diabetic retinopathy dataset. This method cuts down on training time and boosts performance, especially when there isn't much medical data to work with. For the best learning, only the deeper layers of the pretrained networks are unfrozen during training. This lets the models adapt to features that are specific to the domain while keeping general visual representations. The Adam optimizer is used to train the models, and a controlled learning rate is used to make sure that convergence is stable. Regularization methods like dropout [13] and early stopping are used to stop overfitting. Dropout layers randomly turn off neurons while training, which makes the model less dependent on certain features and better at generalizing [13]. Early stopping monitors validation loss and halts training when performance no longer improves, ensuring that the model retains the best learned weights. The dataset is divided into training, validation, and testing sets to enable proper evaluation of model performance. Multiple evaluation metrics, including accuracy, precision, recall, and F1-score, are used to assess the effectiveness of the models, providing a comprehensive understanding of their performance across different classes.

IV. EXPERIMENTAL SETUP

The experiment is set up in the Google Colab platform equipped with an NVIDIA T4 GPU to ensure efficient training of deep learning models. The dataset was divided into training, validation, and testing subsets in a 70:15:15 ratio to ensure proper model evaluation and prevent data leakage. All input images were resized to 224×224 pixels and processed in batches of size 32. The models were trained for a maximum of 25 epochs, with early stopping applied to prevent overfitting and retain the best-performing model based on validation loss. The Adam optimizer was used for training, with a carefully tuned learning rate to ensure stable convergence. To ensure a fair comparison, all models were trained under identical experimental conditions, including the same data

TABLE I
PERFORMANCE COMPARISON OF DIFFERENT DEEP LEARNING MODELS

Model	Training Acc.	Val. Acc.	Testing Acc.
EfficientNetB0	88.34	83.70	81.36
ResNet50	85.43	65.62	67.02
MobileNet	73.31	48.52	48.20
Hybrid1	95.35	87.13	81.36
Hybrid2	93.38	87.50	78.67

splits, preprocessing steps, and evaluation metrics. Model performance was assessed using multiple evaluation metrics, including accuracy, precision, recall, F1-score, and confusion matrix analysis.

V. RESULTS AND DISCUSSION

A. Performance Comparison

Table I presents the training, validation, and testing accuracy of all five models evaluated in this study. Among the individual architectures, EfficientNetB0 achieves the highest testing accuracy of 81.36%, demonstrating strong generalization from training to unseen data. ResNet50 attains a training accuracy of 85.43% but exhibits a notable decline in validation and testing accuracy (65.62% and 67.02% respectively), suggesting overfitting. MobileNetV2 records the lowest performance across all splits, with a testing accuracy of 48.20%, which may be attributed to its lightweight architecture being insufficient for the complexity of multi-class retinopathy classification. Among the hybrid architectures, Hybrid1 achieves the highest validation accuracy of 87.13% and Hybrid2 achieves 87.50%, both outperforming individual models in validation. However, both hybrid models show a larger gap between validation and testing accuracy compared to EfficientNetB0, indicating relatively weaker generalization despite higher feature extraction capability.

B. Evaluation Metrics Comparison

To provide a comprehensive evaluation beyond accuracy — precision, recall, and F1-score were analyzed for all models. Table II presents the weighted average metrics for all models. It is observed that EfficientNet achieves the most balanced performance across all evaluation metrics, particularly in terms of Precision, F1-score, indicating its effectiveness in handling class imbalance. While the proposed hybrid model demonstrates competitive precision, a reduction in recall and F1-score is observed, particularly for minority classes such as severe and proliferative DR. This suggests that although the hybrid architecture enhances accuracy and feature representation, it may exhibit reduced generalization compared to EfficientNet.

C. Confusion Matrix Analysis

Fig. 4 represents the confusion matrix for the diabetic retinopathy implemented using EfficientNetB0. The diagonal elements represent the correct predictions, while the non diagonal elements represent the cases where the model confuses one class with another. It is observed from the confusion

TABLE II
EVALUATION METRICS COMPARISON OF DIFFERENT DEEP LEARNING MODELS

Model	Precision	Recall	F1-Score
EfficientNetB0	0.86	0.84	0.86
ResNet50	0.55	0.56	0.50
MobileNet	0.34	0.50	0.32
Hybrid1	0.79	0.79	0.77
Hybrid2	0.79	0.79	0.79

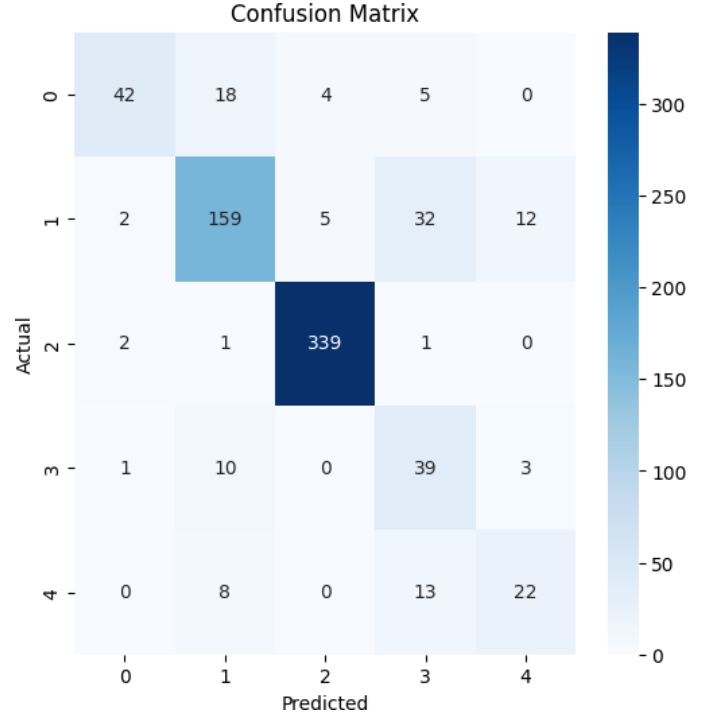


Fig. 4. Confusion matrix of the most efficient model among compared i.e., EfficientNetB0.

matrix that the model achieves strong performance for the “Moderate” class, with a high number of correct predictions, indicating effective learning of retinal features associated with this stage. Similarly, the Mild class demonstrates reasonably good classification performance. However, the “No DR” class shows minor confusion with early stage DR categories.

D. Discussion

The above results show that different deep learning architectures exhibit different strengths in diabetic retinopathy detection and classification. Among the evaluated models based on the above sections, we can say that EfficientNet achieves a higher generalization performance, as shown by its higher test accuracy and evaluation metrics.

A key observation from the comparison analysis is, despite having a slightly higher accuracy of the hybrid model over the EfficientNet, the overall performance based on the evaluation metrics clearly showcases the inability of generalizability of the hybrid model compared to EfficientNet.

The improved validation performance of the hybrid model is achieved by the integration of two fine image analyzing models, i.e., EfficientNet and ResNet, which enables feature extraction at a very fine grained level. The hybrid architectures enhance the feature extraction, whereas EfficientNet displays superior performance and stability.

In the context of medical diagnosis, where reliability and precision are critical, models with higher generalization capability are most suitable for real-world applications.

VI. CONCLUSION

This paper has demonstrated a comprehensive approach for multi-class diabetic retinopathy classification using deep learning techniques. A comparative analysis of various CNNs architectures was conducted and the various architectures include EfficientNetB0, ResNet50, and MobileNetV2. In addition, a hybrid EfficientNet – ResNet model was also compared with the individual CNN architectures. The hybrid model is integrated with an attention mechanism to enhance the feature extraction of the fundus images. The results demonstrate that while the hybrid model achieves improved accuracy due to its ability to extract high level features, whereas EfficientNet displays superior performance and exceptional overall generalization. From the findings, the higher test accuracy is also achieved by EfficientNet only, which is a clear indicator of the suitability for real-world applications. Finally, our findings show that, EfficientNetB0 is identified as the most suitable model for practical deployment due to its stable performance and strong generalization ability.

VII. FUTURE WORK

While the study demonstrates promising results and clear comparison analysis for multi-class diabetic retinopathy classification using various architectures, however various aspects for future research can be explored. One of the main aspects that can be improved is model generalization across different populations and imaging conditions. Integrating data augmentation techniques can further improve the robustness and reduce overfitting. Additionally, emerging architectures like Vision Transformers (ViTs) and hybrid CNN transformers can also be explored which might offer significant improvements in diabetic retinopathy detection. In addition, integrating visualization explanation techniques like saliency maps or Gradient-weighted Class Activation Mapping (Grad-CAM) can also be incorporated into the study which can improve the model predictions significantly.

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